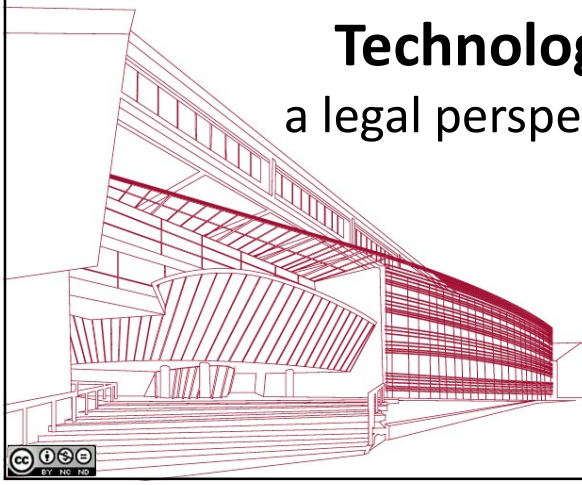
KARL-FRANZENS-UNIVERSITÄT GRAZ
UNIVERSITY OF GRAZ

Cryptoassets and Blockchain Technology

a legal perspective



Univ.-Prof. Dr. Tina Ehrke-Rabel

Institut für Finanzrecht
Karl-Franzens-Universität Graz

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What do you know about cryptoassets?

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- https://finance.ec.europa.eu/digital-finance/crypto-assets_en
- https://artificialintelligenceact.eu/high-level-summary/
- EU AI Act
- https://www.ecb.europa.eu/press/key/date/2026/html/ecb.sp260324~66f71f7577.en.html

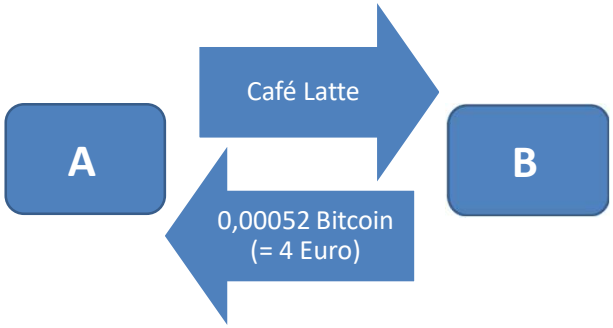
Definition of crypto-asset

A digital representation of value or a right that can be transferred or stored electronically using distributed ledger technology or similar technology.

(definition acc. to the Anti-Money-Laundering Directive)

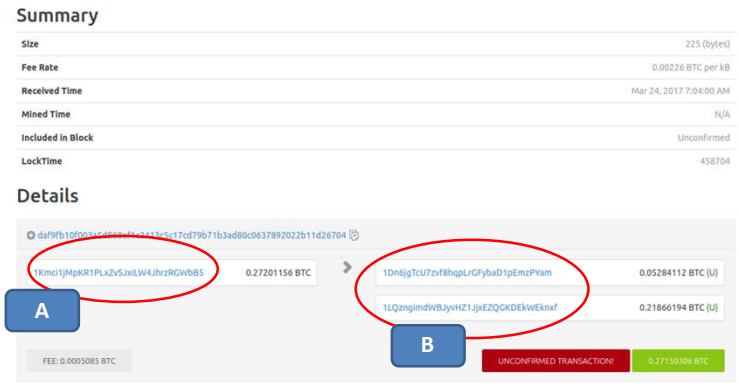
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Bitcoin-Blockchain



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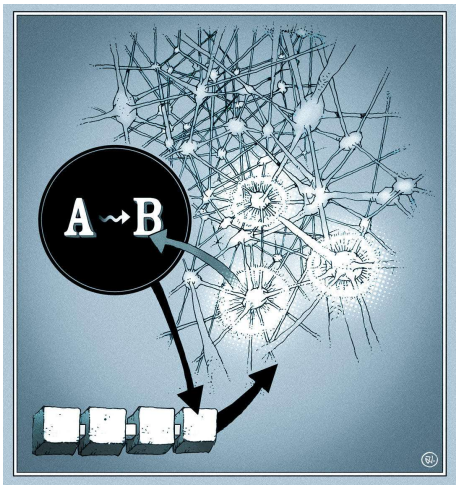
Bitcoin-Blockchain



- 0,00052 Bitcoin = ca 4,00 Euro

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Bitcoin-Blockchain



Graphik: *Jörg Vogeltanz*

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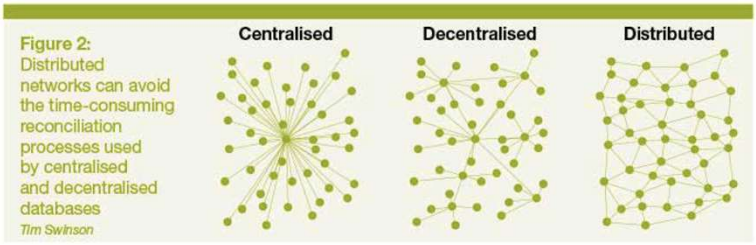
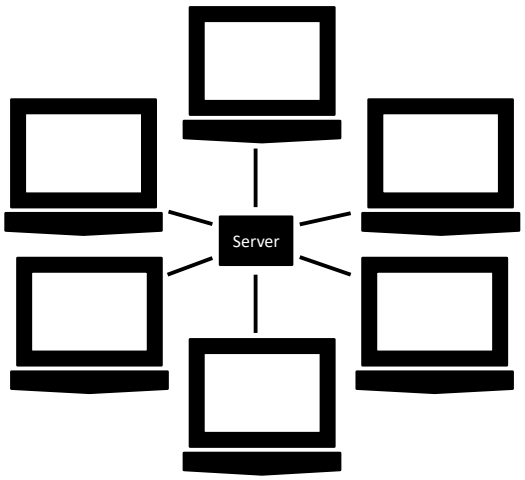


Figure 2: Distributed networks can avoid the time-consuming reconciliation processes used by centralised and decentralised databases
Tim Swinson

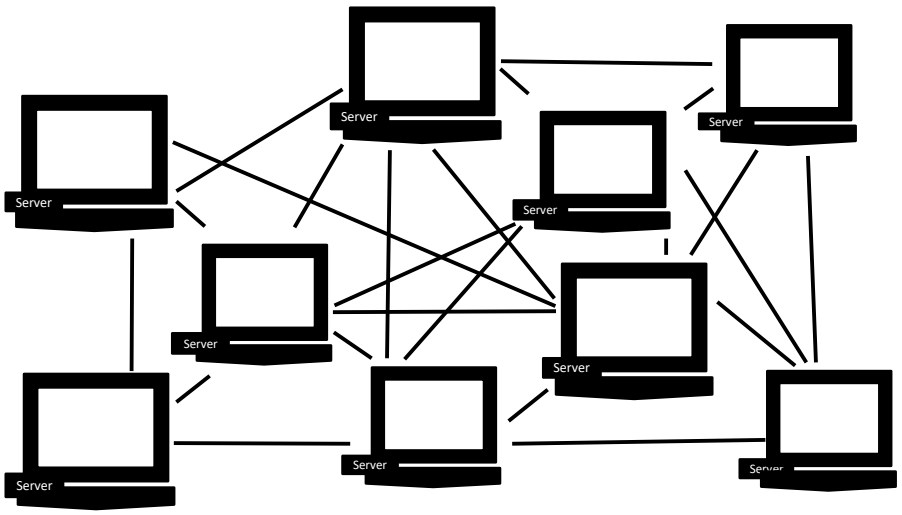
Centralized system



9

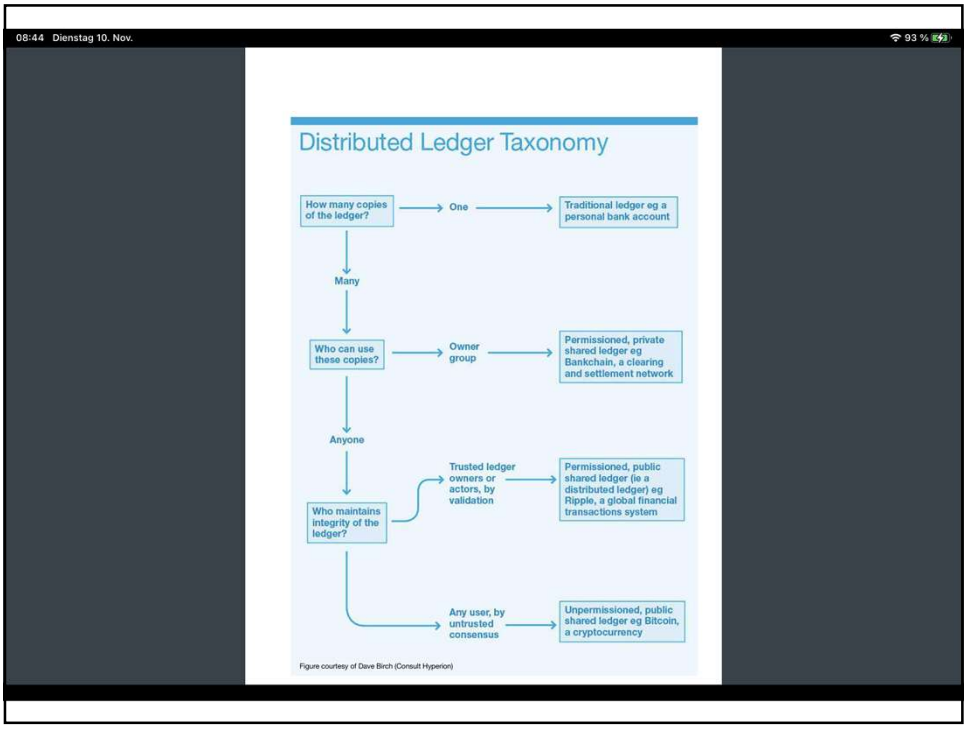
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Decentralised system



10

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08:44 Dienstag 10. Nov.
93 %
Distributed Ledger Taxonomy
How many copies of the ledger?
One
Traditional ledger eg a personal bank account
Many
Who can use these copies?
Owner group
Permissioned, private shared ledger eg Bankchain, a clearing and settlement network
Anyone
Who maintains integrity of the ledger?
Trusted ledger owners or actors, by validation
Permissioned, public shared ledger (ie a distributed ledger) eg Ripple, a global financial transactions system
Any user, by untrusted consensus
Unpermissioned, public shared ledger eg Bitcoin, a cryptocurrency
Figure courtesy of Dave Birch (Consult Hyperion)

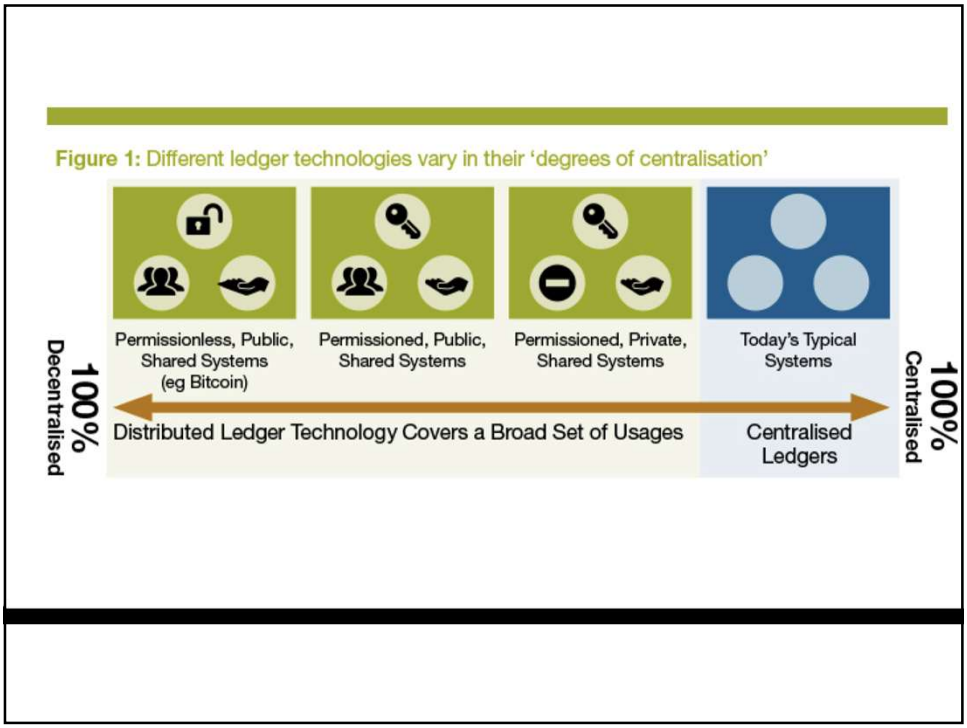
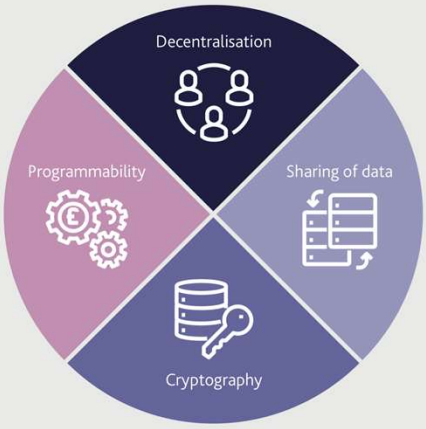

Figure 1: Different ledger technologies vary in their 'degrees of centralisation'
Permissionless, Public, Shared Systems (eg Bitcoin)
Permissioned, Public, Shared Systems
Permissioned, Private, Shared Systems
Today's Typical Systems
100% Decentralised
Distributed Ledger Technology Covers a Broad Set of Usages
Centralised Ledgers
100% Centralised

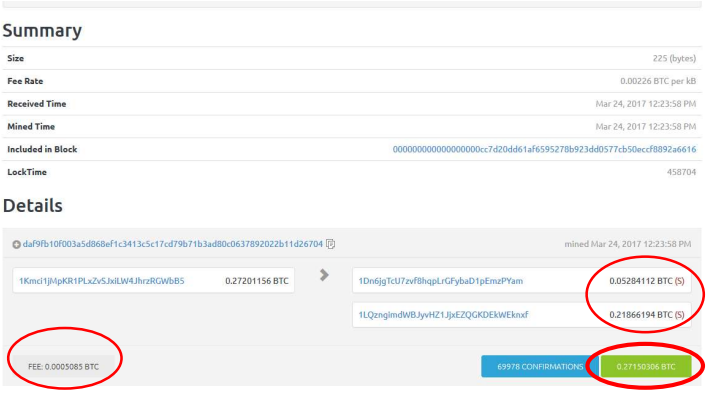
Figure 6.1 Elements of DLT

Each of these elements can potentially be adopted independently of each other



(1) Building blocks: the useful elements of blockchain, Simon Scorer (2019).

Bitcoin-Mining



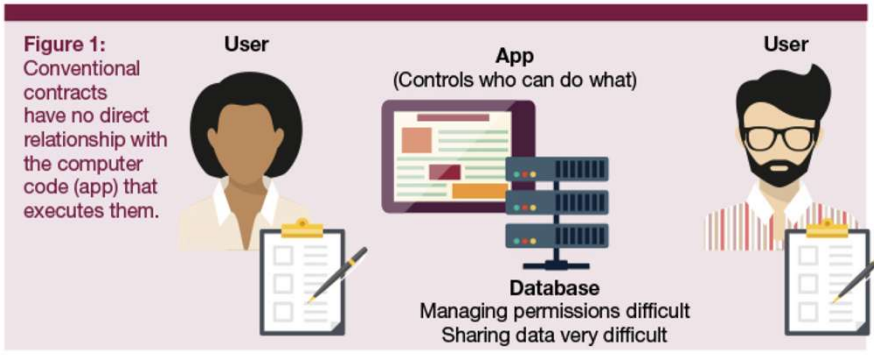
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Crypto-Trading

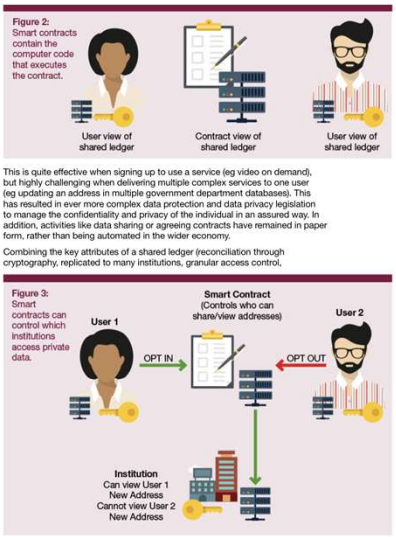
https://www.coindesk.com/price/bitcoin/

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Smart Contracts



Smart Contracts

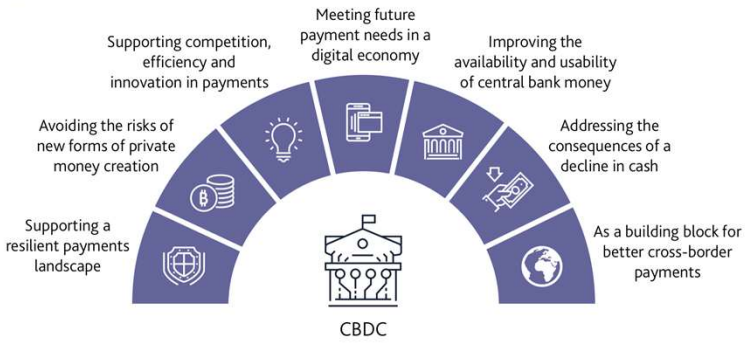


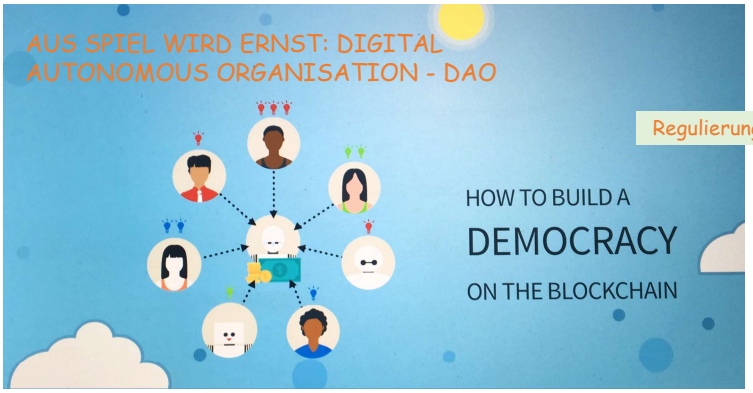
This is quite effective when signing up to use a service (eg video on demand), but highly challenging when delivering multiple complex services to one user (eg updating an address in multiple government department databases). This has resulted in ever more complex data protection and data privacy legislation to manage the confidentiality and privacy of the individual in an assured way. In addition, activities like data sharing or agreeing contracts have remained in paper form, rather than being automated in the wider economy.

Combining the key attributes of a shared ledger (reconciliation through cryptography, replicated to many institutions, granular access control,

https://www.europa.eu/euro/digital_euro/timeline/html/index.en.html

Figure 2.1 Opportunities for CBDC to support monetary and financial stability





Regulierungsbedarf

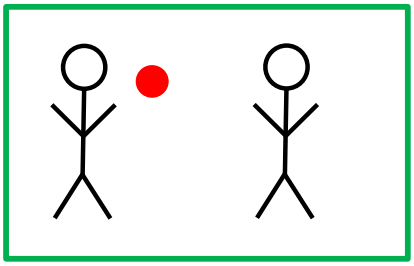
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Do we need to regulate crypto-assets

- ✧ Virtual space without any harm for anyone: no need for regulation
- ✧ Need for regulation when impact in real world

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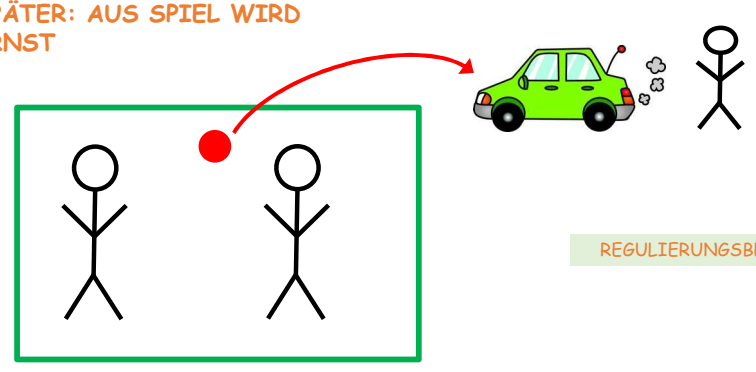
BITCOIN: AM ANFANG WAR DAS SPIEL .....



FREIHEIT

KEIN REGULIERUNGSBEDARF

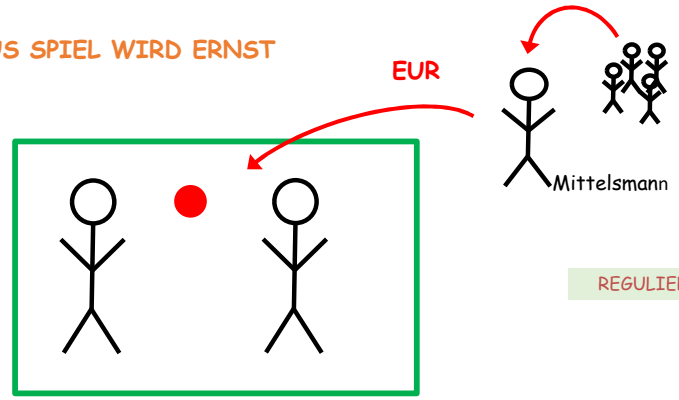
SPÄTER: AUS SPIEL WIRD ERNST



REGULIERUNGSBEDARF

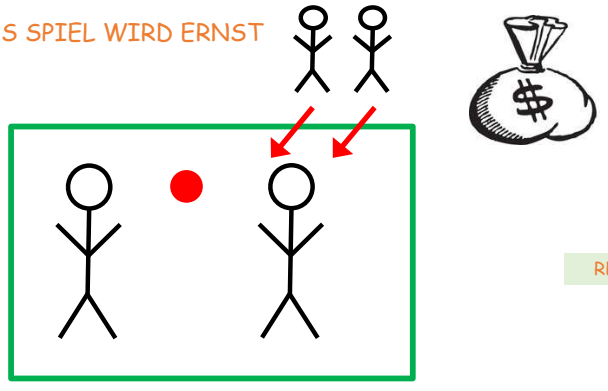
AUS SPIEL WIRD ERNST

EUR



REGULIERUNGSBEDARF

AUS SPIEL WIRD ERNST



REGULIERUNGSBEDARF

What is a cryptoasset?

- A means of payment?
- An intangible good?
- Nothing?

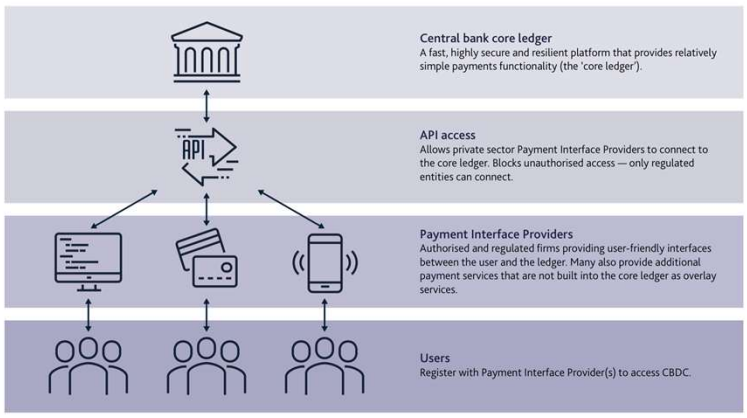
25

Digitales Zentralbankgeld
CBDC (Central Bank Digital Digital Currency)

https://cbdctracker.org

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Figure 4.1 Platform model of CBDC



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Unter Mitarbeit von
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Mag.a Sarah Domes-Hohl
Mag.a Nora Schreier

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